

RECAP: 1) Random Processes:

- Definition: $\{X_t, t \in T\}$, collection of RVs on the same $(\Omega, \mathcal{F}, P)$
- T : index set; if T countably infinite: discrete parameter process
if T uncountably infinite: continuous parameter process
- Three ways to look at a Random Process:
 - for a fixed $t \in T$, $X_t: \Omega \rightarrow \mathbb{R}$ a random variable
 - for a fixed $\omega \in \Omega$, $X(\omega): T \rightarrow \mathbb{R}$, a sample path
 - as a function of two variables $X(t, \omega): T \times \Omega \rightarrow \mathbb{R}$
- Finite dimensional distribution: $F_{t_1, t_2, \dots, t_m}(x_1, x_2, \dots, x_m)$ - Consistency.
- Properties: Mean: $\mu_X(t)$, Autocorrelation: $R_X(t, s)$, Autocovariance: $C_X(t, s)$

2) Stationarity: $\{X_t: t \in T\}$ a random process w.r.t $(\Omega, \mathcal{F}, P)$ a) Strict Sense: $F_{t_1, \dots, t_n}(x_1, \dots, x_n) = F_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$, $\forall n \in \mathbb{N}$ and $\forall \{t_1, \dots, t_n\} \subset T$ Translation-invarianceb) N^{th} order: $F_{t_1, \dots, t_n}(x_1, \dots, x_n) = F_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$ $\forall n \leq N, n \in \mathbb{N}$
 $\forall \{t_1, \dots, t_n\} \subset T$ c) Wide Sense: $\mu_X(t)$ is a constant
 $R_X(s, t)$ is a function of $s-t$ only, $E[X_t^2] < \infty \forall t \in T$.PLAN: 1) IID Process: Bernoulli Process.

2) Arrival and Renewal processes.

3) Poisson Process.

Reference: Bruce Hajek, "Random Processes for Engineers".
R. G. Gallager, "Stochastic Processes Theory for Applications"

IID Processes

Fix a probability space $(\Omega, \mathcal{F}, P)$. Let $\{X_t : t \in \mathcal{T}\}$ be a random process defined w.r.t $\mathcal{F}$

Defn: $\{X_t : t \in \mathcal{T}\}$ is said to be an **IID Process** with the **common CDF F** if for any $n \in \mathbb{N}$ and $\{t_1, t_2, \dots, t_n\} \subset \mathcal{T}$, $F_{t_1, t_2, \dots, t_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n F(x_i)$, $x_1, \dots, x_n \in \mathbb{R}$

i.e. For any $n \in \mathbb{N}$ and $\{t_1, t_2, \dots, t_n\} \subset \mathcal{T}$, $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ are i.i.d random variables

- If $\{X_t : t \in \mathcal{T}\}$ is an IID process:
- the FDDs of $\{X_t : t \in \mathcal{T}\}$ are consistent
 - $\{X_t : t \in \mathcal{T}\}$ is strictly stationary

Every IID Process is Stationary!

Example: **IID Process**.

Consider a prob. space $(\Omega, \mathcal{F}, P)$

$X_1, X_2, X_3, \dots$ defined w.r.t $\mathcal{F}$, i.i.d with $P(X_i = 1) = p$ and $P(X_i = -1) = 1-p$

$\{X_i : i \in \mathbb{N}\}$ is an i.i.d Bernoulli process.

$$F_{t_1, t_2, \dots, t_n}(x_1, \dots, x_n) = \prod_{i=1}^n F_{t_i}(x_i) \quad \text{Are they consistent?}$$

$$\mu_X(t) =$$

$$R_X(t, s) =$$

$$C_X(t, s) =$$

Let X_0 be an integer valued random variable on the same prob. space independent

of $X_1, X_2, \dots$

$$\text{We define } Z_n = \sum_{i=0}^n X_i$$

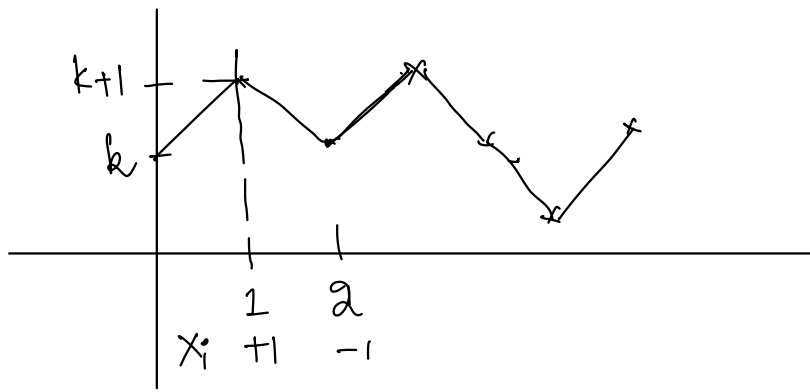
$\{Z_n : n \geq 0\}$ is a random walk

What is $E[Z_n | X_0 = k]$?

$$\text{Var}(Z_n | X_0 = k) = ?$$

$$Z_n = X_0 + X_1 + \dots + X_n$$

Homework



Gambler's Ruin: Gambler's initial wealth is k , $k \geq 0$
 Z_n is the number of units of money he has at time n .

Let the gambler have a goal of accumulating b units of money, $b \geq k$
 While the random walk $\{Z_n: n \geq 0\}$ continues forever, we are only interested in it until it hits either 0 (gambler is ruined) or b (the gambler is successful).

Let S_b denote the event that the gambler is successful, meaning the random walk reaches b before reaching 0.

Gambler's success prob. $s_k \triangleq P(S_b | X_0 = k)$ — success probability for a gambler with initial wealth k and target wealth b .
 $s_0 = 0$, $s_b = 1$ (Boundary Conditions)

$$s_k = P\{X_1 = 1\} P\{S_b | X_1 = 1, X_0 = k\} + P\{X_1 = -1\} P\{S_b | X_1 = -1, X_0 = k\}$$

$$= p s_{k+1} + (1-p) s_{k-1} \quad \text{for } 1 \leq k \leq b-1$$

This yields $(b-1)$ linear equations in $(b-1)$ unknowns $s_1, \dots, s_{b-1}$

When $p = 1/2$, $s_k = \frac{1}{2} \{s_{k+1} + s_{k-1}\}$, $1 \leq k \leq b-1$

With boundary conditions $s_0 = 0$ and $s_b = 1$, we get $s_k = k/b$.

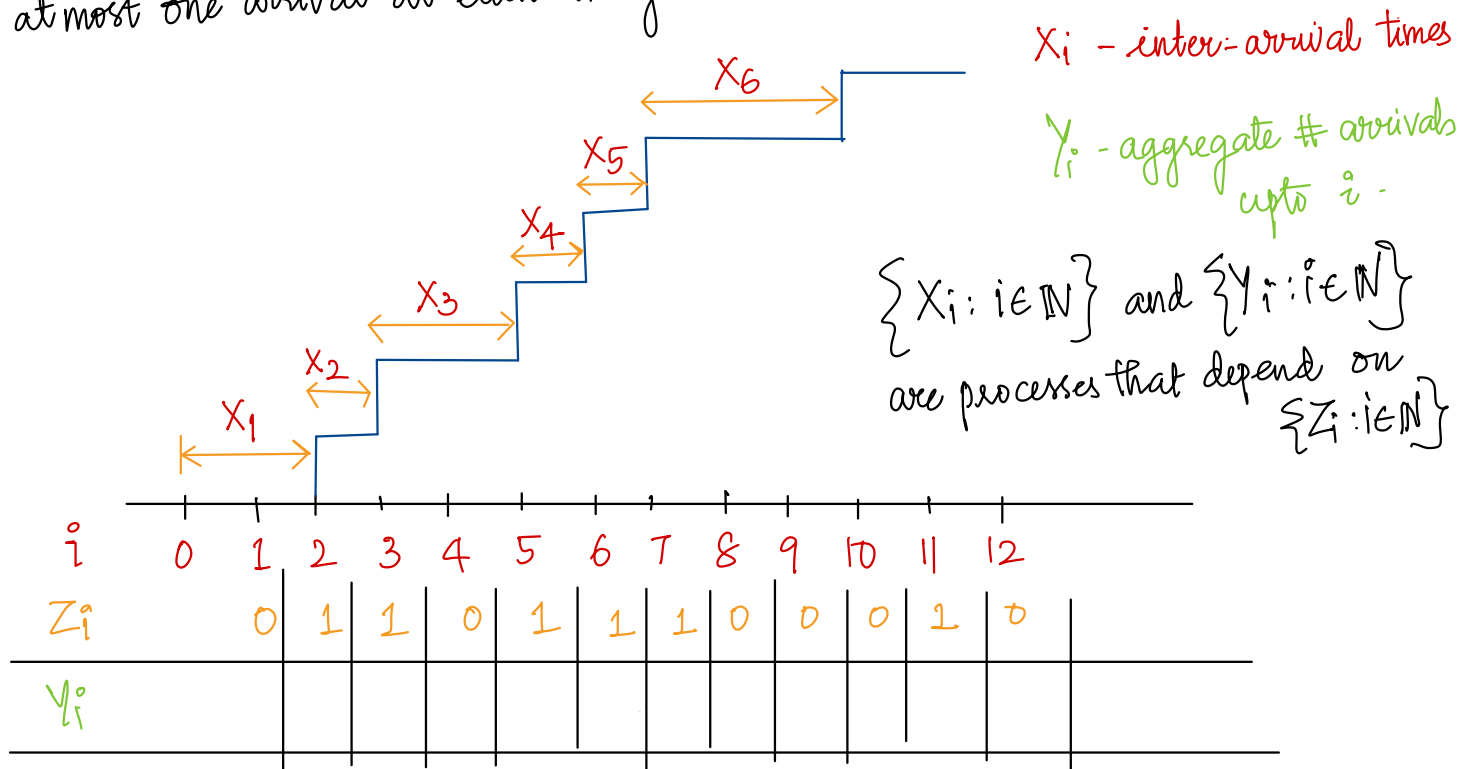
Note that when the gambler stops playing, he has b units of money with prob. k/b and 0 units otherwise $\therefore$ His expected wealth is $b \cdot \frac{k}{b} + 0 \cdot (1 - \frac{k}{b}) = \underline{k}$ (his initial wealth)

Bernoulli Process :

The Bernoulli process is a sequence $Z_1, Z_2, \dots$ of iid binary RVs.

Let $p = \Pr(Z_i = 1)$ and $q = 1 - p = \Pr(Z_i = 0)$

Bernoulli Process : evolving in discrete time with the event $\{Z_i = 1\}$ representing a customer arriving at instant i and $\{Z_i = 0\} \Rightarrow$ no arrival.
i.e., at most one arrival at each integer instant.



$$X_1 = 1 \text{ iff } Z_1 = 1 \Rightarrow P_{X_1}(1) = p.$$

$$X_1 = 2 \text{ iff } Z_1 = 0 \text{ and } Z_2 = 1 \Rightarrow P_{X_1}(2) = (1-p)p.$$

$$X_1 \sim \text{Geo}(p)$$

$$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Geo}(p)$$

$$X_2 \sim \text{Geo}(p)$$

$\therefore$ Sequence of inter-arrival times is also an IID process with geometric distribution.

A sample path of $\{X_i : i \in \mathbb{N}\}$ determines a sample path of $\{Z_i, i \in \mathbb{N}\}$.

$\Rightarrow$ Bernoulli process can be characterised by a sequence of geometric RVs.

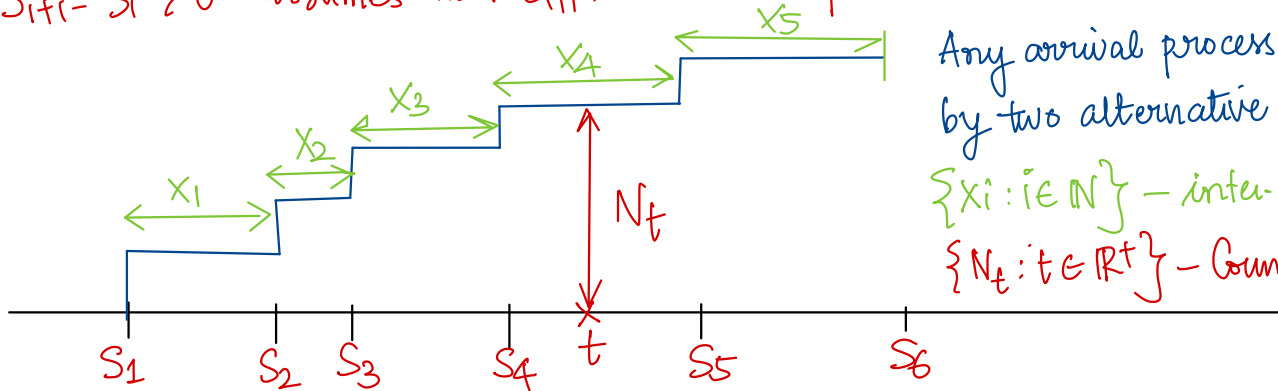
$\{Y_n: n \in \mathbb{N}\}$ — where $Y_n = \sum_{i=1}^n Z_i$ — number of arrivals upto and including the time instant n .

Qn: What is the distribution of Y_n ?

Arrival process: A sequence of increasing RVs $0 < S_1 < S_2 < \dots$, where $S_i < S_{i+1}$
 $\Rightarrow S_{i+1} - S_i$ is a positive RV X s.t. $F_X(0) = 0$. The RVs $S_1, S_2, \dots$ are called arrival instants/epochs and represent successive times at which some random repeating phenomenon occurs (like arrivals, departures, etc).

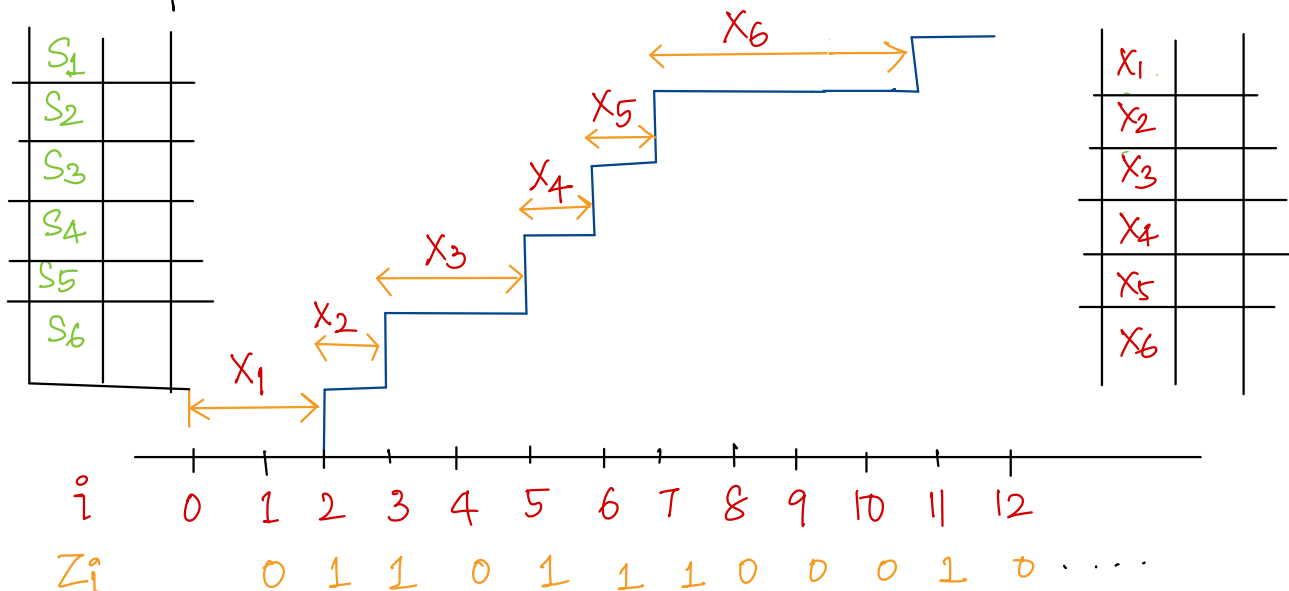
We saw Bernoulli process $\{Z_i: i \in \mathbb{N}\}$ modeling arrivals at integer values.
 $Z_i = 1 \Rightarrow$ there is an arrival in the epoch i , $i \in \mathbb{N}$.

$S_{i+1} - S_i > 0$ assumes that $S_{i+1} \neq S_i \Rightarrow$ Multiple arrivals cannot occur simultaneously.



Any arrival process can be specified by two alternative stochastic processes:
 $\{X_i: i \in \mathbb{N}\}$ — inter-arrival times
 $\{N_t: t \in \mathbb{R}^+\}$ — Counting process

For this sample path of the Bernoulli process, what is the sample path of the corresp. arrival process and the inter-arrival times?



To fully specify the arrival process, we need to specify the FDDs of the process $\{S_n : n \in \mathbb{N}\}$

Alternate characterizations of the Arrival Process:

i) Sequence of inter-arrival times : $\{X_i : i \in \mathbb{N}\}$: Continuous valued discrete parameter process.

$$X_1 = S_1$$

X_i s are positive RVs

$$X_i = S_i - S_{i-1}, i > 1$$

$$S_n = \sum_{i=1}^n X_i \quad \left\{ \text{Arrival epochs in terms of inter-arrival times} \right\}.$$

Thus, the FDDs of the process $\{X_n : n \in \mathbb{N}\}$ also completely specify the arrival process. In most cases of interest, the inter-arrival times are IID and hence the arrival process is specified using the FDDs of the inter-arrival times.

ii) The Counting Process : $\{N_t : t \in \mathbb{R}^+\}$: (Discrete-valued Continuous parameter process)

N_t - Aggregate number of arrivals in the interval $[0, t]$

Assumption : Arrivals at strictly positive values of t i.e. $S_1 > 0$

$N_0 \triangleq 0$ with prob. 1.

$$N_t = n \text{ for } S_n \leq t \leq S_{n+1}.$$

$$N(t_2) \geq N(t_1) \text{ for } 0 \leq t_1 \leq t_2$$

The arrival process and the counting process are related as :

$$\{S_n \leq t\} = \{N_t \geq n\}.$$

$$\text{(OR) equivalently } \{S_n > t\} = \{N_t < n\}$$

Example: Processes in a single station queue.

Consider the CCD outlet in B-Block. There is only one server.

Customers arrive at random times and "suppose" they wait in line in FCFS order.

$\{A_n: n \in \mathbb{N}\}$: Arrival epochs. $\{D_n: n \in \mathbb{N}\}$: Departure epochs.

$\{S_t^A: t \in \mathbb{R}^+\}$ = Aggregate arrivals in $(0, t]$

$\{S_t^D: t \in \mathbb{R}^+\}$ = Aggregate departures in $(0, t]$.

$\{X_t: t \in \mathbb{R}^+\}$: Queue length process = # customers in the system at time t .

$\{W_n: n \in \mathbb{N}\}$: Sojourn times of the customers. (Amount of time n^{th} customer spends in the system from the instant she arrives until the instant she leaves.)

Renewal Process: A renewal process is an arrival process for which the sequence of inter-arrival times is a sequence of positive IID random variables.

Example: Arrival process that is not a renewal process.

1) You toss a fair coin. If H, interarrival times $X_i = 1 \ \forall i \in \mathbb{N}$

If T, Bernoulli arrival process. i.e., arrivals occur independently at every discrete time instant with prob $1/2$. Hence $X_1, X_2, \dots | T \stackrel{\text{iid}}{\sim} \text{Geo}(1/2)$ with support $1, 2, \dots$

Thus $P(X_i > t | T) = (1/2)^t$

If H, $P(X_i > s | H) = 0$ for any $s \geq 1$

Observing $X_1 > 1$ tells us that the coin flip resulted in Tails. $P(T | X_1 > 1) = 1$

$P(X_2 > t | X_1 > s) = (1/2)^t \quad s \geq 1, t \geq 0$

Compare with the unconditional distribution of X_2 . For $t \geq 1$

$P(X_2 > t) = P(X_2 > t | H) \cdot P(H) + P(X_2 > t | T) \cdot P(T) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot (1/2)^t$

$$P(X_2 > t) = \left(\frac{1}{2}\right)^{t+1}$$

$$\therefore P(X_2 > t | X_1 > s) = \left(\frac{1}{2}\right)^t \neq \left(\frac{1}{2}\right)^{t+1} = P(X_2 > t)$$

$\therefore X_1$ and X_2 are not independent.

Intuitively, if you observe $X_1 > 1$, we immediately know the process is iid $\text{Ber}(1/2)$.

Defn. A **Poisson process** is a renewal process in which the inter-arrival times have an **exponential CDF**, i.e. for some real $\lambda > 0$, each X_i in $\{X_i : i \in \mathbb{N}\}$ has the pdf $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

λ : Rate of the process.

Why is it called a Poisson process?

Ans: Counting Process N_t has a Poisson distribution.

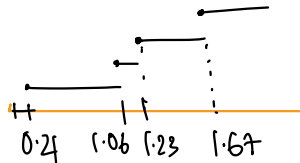
Example: Let the arrival rate be $\lambda = 2$ arrivals per minute

Description 1: Using inter-arrival times.

$X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \exp(\lambda)$

$X_i(\text{min})$ Arrival $S_i(\text{min})$

1)	0.21	0.21
2)	0.85	1.06
3)	0.17	1.23
4)	0.44	1.67



t	0	0.5	1	1.5	2
N_t	0	1	1	3	4

Description 2: Using Poisson Counts.

Instead of generating arrival times, we find the prob of seeing a certain # of arrivals at a given time:

$$X \sim \text{Pos}(\lambda) \Rightarrow$$

$$N_t \sim \text{Poisson}(\lambda t)$$

$$N_2 \sim \text{Poisson}(4)$$

$$P_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$\mathbb{E}[N_2] = 4$$

Poisson processes constitute the simplest and the most widely-used class of renewal processes. Why? Because of the **MEMORYLESS** property of the exponential distribution.